Efe-Oghene Erhie

MEng candidate at Waterloo, moving from mechanical engineering into robotics software.

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Scan for projects & live demos

EDUCATION

MEng, Mechanical & Mechatronics Engineering
University of Waterloo — Sep 2026 – present
Co-operative, course-based

BSc, Mechanical Engineering
University of Manitoba — Graduated Jun 2022
GPA 4.36 / 4.50 · Best Graduating Student, Department of Mechanical Engineering

EXPERIENCE

Antec Controls
Intermediate Manufacturing Specialist — Jan 2026 – Aug 2026
Manufacturing Specialist — Nov 2022 – Jan 2026

- Led the project that established manufacturing at a new site, meeting country-of-origin and domestic-content requirements the existing facility could not satisfy.
- Planned the setup end to end and executed it on site, taking the new space to a peak of 200 valves in a single day within its first week.
- Operated and maintained automated test equipment for airflow testing of valves used in critical environments.
- Owned technical documentation and calibration procedures maintaining NVLAP accreditation for airflow measurement systems.
- Supported planning and execution of a production facility relocation into a space twice the size of the previous one.
- Provided strategic support implementing a new ERP system (IFS Cloud) integrating manufacturing and inventory processes.

PepsiCo Beverages Canada
Operations Leadership Trainee — Aug 2022 – Oct 2022

- Led a project to improve reliability of critical equipment at the beverage production plant.
- Coordinated equipment contractors to maximise the value of on-site visits.
- Analysed and improved the changeover process on key machines of the bottling line.

Academic Learning Centre, University of Manitoba
Content / Study Skills Tutor — Aug 2021 – May 2022

- Tutored engineering and science courses including thermodynamics, fluid mechanics, and CAD.
- Coached students on study technique and time management to build independent learning capability.

PepsiCo Beverages Canada
Production Shift Supervisor (Co-op) — Jan 2021 – Sep 2021

- Directed a production team against cost, line efficiency, waste, safety and productivity standards.
- Oversaw facility equipment reliability to minimise downtime.
- Built front-line capability through training and communication.

TECHNICAL SKILLS

Languages	Python, C++, Rust, JavaScript
Simulation & numerical	NumPy vectorisation, spatial partitioning, control-loop tuning, cross-implementation performance profiling
Systems	Rust/WebAssembly toolchain, Tauri, event-sourced data models, static-site architecture
Engineering	SolidWorks, AutoCAD, Fusion 360, GD&T, automated test equipment, calibration procedure authorship, NVLAP-compliant documentation, ERP systems (IFS Cloud)

SELECTED PROJECTS

Full write-ups and runnable demos at efeerhie.com/projects.

Boids flocking simulation

Python · NumPy · Rust · WebAssembly

A flocking model built three times over to study the same system from different angles: a scalar Python reference, a vectorised NumPy rewrite, and a Rust core compiled to WebAssembly. Covers control-loop stability under changing gains, and the move from an $O(n^2)$ neighbour scan to a spatial grid.

omni-me

Rust · Tauri · Dioxus

A cross-platform personal-data application. Event-sourced transaction model with multi-source reconciliation, recurring-transaction detection, and multi-currency account handling.

mylearnbase

Zola · Tera · JavaScript

A static portfolio and learning journal at mylearnbase.com. Build-time TF-IDF for content-based related posts, in-browser search over a pre-built index, and self-contained interactive demos embedded in articles.

AWARDS

Best Graduating Student, Department of Mechanical Engineering	2022
Dean's Honour List, Faculty of Engineering	2018 – 2022
UMSU Scholarship	2019 – 2021
International Undergraduate Student Scholarship	2019 – 2021
Grettir Eggertson Memorial Scholarship	2019 – 2021

INTERESTS

Archery · Dragon boat · University of Waterloo clubs